What Makes Audio Latents Mixing-Equivariant? A Controlled Study of Explicit Supervision

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***Abstract*—Mixing is a linear operation on waveforms, so a latent space that respects it, one in which interpolating two codes and decoding reproduces the waveform mix, is an *equivariant* representation. Neural audio autoencoders do not learn this structure on their own. Recent work induces it in a consistency autoencoder by decoding averaged latents against the true mix; we ask which part of such supervision is actually responsible, whether the decoder class limits what it delivers, and whether the metrics used to certify it measure it at all. We answer with a controlled study in which each experiment intervenes on a single factor while holding architecture, data, initialization, and schedule fixed. An explicit *decode-mixing loss*, which requires decoded latent interpolations to match true waveform mixes, at the cost of one extra decoder pass, is sufficient: on a waveform GAN autoencoder it makes the latent space mixing-equivariant *at no measured reconstruction cost*, and at $2\times$ tighter compression the same holds against a matched from-scratch control. Raw latent subtraction of a stem improves from $+3.4$ to $+5.7$ dB SI-SDR with the loss, but this advantage is a latent-offset effect: adding the encoded silence $f(0)$, exact for affine maps, brings every model, with or without the loss, to within 0.5 dB of its reconstruction ceiling on all 49 test tracks. What the loss buys is convex equivariance; subtraction needs only the origin. All comparisons are single-run, matched at one initialization. The interventions also identify what is *not* responsible: an adversarial term on the mixed path inflates a decode-consistency metric without improving ground-truth equivariance, exposing that metric as a confounded proxy; an encoder-only penalty linearizes the encoder without making the decoder equivariant; and on a consistency-model autoencoder (Music2Latent) a continued-training control shows the loss improves latent geometry, although absolute waveform arithmetic stays poor there: the decoder class sets what linearity can deliver. Finally, the widely used decode-vs-decode linearity metric is shown to be confounded by decoder phase variance, and a ground-truth-referenced variant is reported throughout.**

***Index Terms*—representation learning, equivariant representations, audio autoencoders, latent arithmetic, controlled ablation, source separation**

I. Introduction

Audio mixing is a linear operation: $\mathrm{mix}(x_1, x_2, \alpha) = \alpha x_1 + (1-\alpha)x_2$. For an autoencoder with encoder f and decoder g, we call the latent space *mixing-equivariant* when

$$g(\alpha f(x_1) + (1-\alpha) f(x_2)) \approx \alpha x_1 + (1-\alpha)x_2. \qquad (1)$$

Eq. 1 constrains combinations whose coefficients sum to one. Latent *subtraction*, $g(f(x_{\mathrm{mix}})-f(x_{\mathrm{stem}}))$, uses coefficients $(1, -1)$ and so is not implied by it: an offset $f(x) = x + b$, $g(z) = z - b$ satisfies Eq. 1 exactly yet subtracts to $x_{\mathrm{res}} - b$. Subtraction additionally needs the latent map to have no offset, and we treat it as an empirical question (Section IV-C). If it holds, latent vector arithmetic becomes audio editing, removing, adding, and remixing sources without a dedicated separation or synthesis model. It is also a prerequisite for latent-diffusion systems [1], [2] that compose independently modeled sources. Yet standard autoencoders do not satisfy Eq. 1: their objectives reward reconstruction fidelity, not algebraic structure, and quantized codecs (DAC [3], EnCodec [4]) cannot represent interpolated codes at all.

Torres et al. [5] induce linearity in a consistency autoencoder by decoding the *average* of separately encoded latents and training it to reconstruct the true mix, a supervision principle related to mixup [6]. This leaves open which part of that supervision is responsible for the property, whether the encoder or the decoder must carry it, whether a consistency decoder can turn latent linearity into usable waveform arithmetic, and whether the metrics used to certify it measure it at all. We answer these questions with a controlled study: the same principle written as an explicit loss (Section III) on a waveform GAN autoencoder, whose effect we then isolate by intervening on one factor at a time. Our contributions are:

1) A controlled study of mixed-latent supervision, instantiated as an explicit decode-mixing loss on a waveform GAN autoencoder, where it yields mixing-equivariance *without degrading reconstruction* (Section IV-A); a downstream analysis shows latent stem removal is limited by the latent origin rather than by the loss: adding $f(0)$ brings every model to its reconstruction ceiling (Section IV-C).
2) An interventional ablation isolating the mechanism: the decode-mixing loss alone is sufficient; adding an adversarial term on the mixed path inflates a decode-consistency metric with no ground-truth gain, exposing the metric as a confounded proxy; an encoder-only penalty linearizes the encoder without making the decoder equivariant; and freezing the encoder retains most of the gain, so the effect is carried through the decoder and is largest when both networks adapt.
3) Evidence that the recipe generalizes, each against a matched control: from-scratch training at $2\times$ tighter compression, and a matched re-run on a consistency-model autoencoder with a continued-training control ruling out domain adaptation, where latent geometry im-

proves but waveform-level arithmetic does not become usable. The gain holds on all three test domains.

4) A methodological correction: the widely used decode-vs-decode SI-SDR$_{\mathrm{lin}}$ metric is confounded by decoder phase variance; we report a ground-truth-referenced variant.

II. Related Work

Neural audio codecs. DAC [3], EnCodec [4], and SoundStream [7] pair convolutional encoders with HiFi-GAN decoders [8] and multi-scale discriminators, optimizing reconstruction without latent structure; their residual vector quantization also precludes latent interpolation. RAVE [9] learns a variational latent space but does not target mixing equivariance, and disentanglement regularizers [10] encourage generic smoothness rather than a specific algebraic identity. **Equivariant representations.** Mixing-equivariance is an instance of the symmetry-based view of representation learning, in which a representation is characterized by how it transforms under known operations on the data [11]; convex mixing is an affine combination rather than a group action (it has no inverses), so we borrow the framing, not the group formalism. Verma and Berger [12] build equivariance to audio transformations into the architecture. We instead treat equivariance to the linear mixing operation as a property to be *induced* by supervision, and isolate by intervention which supervision induces it. **Linear audio representations.** Music2Latent [13] is a consistency autoencoder with partial linearity as a byproduct. Torres et al. [5] make it approximately linear by conditioning the decoder on the average of separately encoded latents while denoising the true mix under the unchanged consistency loss (plus a random gain for homogeneity, which we do not study). Our $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$ is the explicit-loss form of that additivity supervision: the same input (averaged latents) and target (true mix), differing in loss family (multi-resolution STFT and mel versus consistency), mixing coefficient ($\alpha \sim U[0,1]$ versus 0.5), and decoder class (deterministic GAN versus consistency model). We add the analysis: which ingredient carries the effect, a matched re-run of the principle on their architecture with a continued-training control (Section IV-B), and ground-truth-referenced metrics. Their decoder-additivity scores (SNR, spectral distance) and separation SI-SDR compare two decodes (decoded latent sum versus the *autoencoded* mix or source), the protocol Section III-B shows to be confounded, and their latent-arithmetic separation is negative on every stem (-0.3 to -1.8 dB SI-SDR), as our phase-coherence finding predicts.

III. Method

A. Decode-mixing loss

Given a batch $\{x_i\}$ we form pairs (x_i, x_j) with a fixed-point-free permutation and draw $\alpha \sim U[0,1]$. Let $\bar{x} = \alpha x_i + (1-\alpha)x_j$ be the waveform mix and $\bar{z} = \alpha f(x_i) + (1-\alpha) f(x_j)$ the latent interpolation. The decode-mixing loss requires the decoded interpolation to match the true mix:

$$\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}} = \mathcal{L}_{\mathrm{recon}}(g(\bar{z}),\, \bar{x})\,, \qquad (2)$$

where $\mathcal{L}_{\mathrm{recon}}$ is the multi-resolution STFT + mel loss also used for reconstruction. Gradients flow through g and back into f via $\bar{z}$, so a single term constrains both networks; it costs one extra decoder pass per step. We also study an *encoder-only* variant that penalizes encoder nonlinearity directly, $\mathcal{L}_{\mathrm{mix}}^{\mathrm{enc}} = \|f(\bar{x}) - \bar{z}\|^2$ (one extra encoder pass, no decoder pass). The total generator objective is $\mathcal{L} = \mathcal{L}_{\mathrm{recon}} + \mathcal{L}_{\mathrm{GAN}} + \lambda\, \mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}} + \lambda_{\ell_2} \|z\|^2$.

B. Metrics

We report reconstruction SI-SDR$_{\mathrm{rec}} = \mathrm{SISDR}(g(f(x)), x)$ and two equivariance views at α=0.5:

- **SI-SDR$_{\mathrm{lin}}$** (decode-vs-decode): $\mathrm{SISDR}(g(\bar{z}), g(f(\bar{x})))$, equivariance relative to the model's *own* reconstruction of the mix.
- **SI-SDR$_{\mathrm{lin}}^{\mathrm{gt}}$** (vs. ground truth): $\mathrm{SISDR}(g(\bar{z}), \bar{x})$, the quantity in Eq. 1 up to scale, comparable across models. SI-SDR fits a gain before scoring, so it does not assess level preservation; we report the fitted gain separately (Section IV-B).

We also report $\ell_{\mathrm{lat}} = \|\bar{z} - f(\bar{x})\|^2 / \|f(\bar{x})\|^2$ (encoder linearity) and MixRate $= \mathcal{L}_{\mathrm{recon}}(g(\bar{z}), \bar{x}) / \mathcal{L}_{\mathrm{recon}}(g(f(\bar{x})), \bar{x})$ (decoder equivariance; < 1 is better than the encode-then-decode path). SI-SDR$_{\mathrm{lin}}$ shares the decode-vs-decode structure of prior work's decoder-additivity metrics (SNR and spectral distance between two decodes [5]), and it is *gameable*: we hypothesize that a decoder driven onto a common realistic manifold raises the mutual SI-SDR of its two decode paths without bringing either closer to $\bar{x}$ (Section IV-A). It is also confounded for stochastic decoders: a consistency-model decoder [13] draws fresh noise per call, so two decodes of the same latent differ in phase and SI-SDR$_{\mathrm{lin}}$ collapses unless the noise is shared. Fig. 1 shows the same Music2Latent checkpoints scoring ≈ -8 dB under independent-noise decoding (the protocol implied by prior reports) and *positive* under shared-noise decoding, a 12 dB swing driven by the eval protocol rather than the model. This shows protocol sensitivity; decoding the same latent twice with independent noise isolates how much of the disagreement arises without any latent arithmetic (Section IV-B). We therefore treat SI-SDR$_{\mathrm{lin}}^{\mathrm{gt}}$ and MixRate as the equivariance metrics of record, and use shared decode noise everywhere.

C. Experimental setup

The primary model is a waveform autoencoder (DAC-style 1-D convolutional encoder, HiFi-GAN decoder [8], combined multi-scale-STFT and multi-period discriminator) operating at 44.1 kHz on 1-second mono chunks, trained on a ~950-hour union of FMA-large, MAESTRO, and MUSDB18-HQ [14].[1] Two compression regimes are used: 7.66× (d=128) and 15.3× (d=64). Mixing losses are added either as a 25k-step fine-tune over a converged baseline (v2.x) or from scratch over 250k steps (v3.x). All runs use a single NVIDIA RTX 5090.

[1] Code, training configurations, pretrained weights, every metric reported here, and audio examples: https://github.com/nurdauletakhanov/musicgen

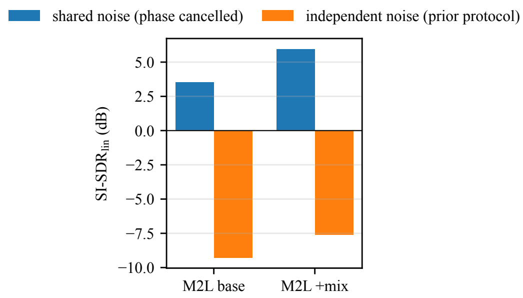

Fig. 1. Decode-noise protocol confound (Music2Latent, MUSDB18, α=0.5). The same checkpoints score deeply negative under independent-noise decoding and positive under shared-noise decoding; the ≈ -8 dB figure reflects decoder noise, consistent with phase variance, not latent linearity.

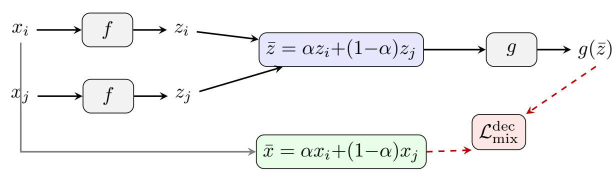

Fig. 2. Decode-mixing loss: encode a pair, interpolate latents, decode, and compare to the true waveform mix. Gradients (dashed) flow through the decoder and back into the encoder via $\bar{z}$.

Unless noted, metrics are computed over the full multi-source test set; the α-sweep (Section IV-D) uses a source-balanced subsample.

IV. Results

A. Which signal causes equivariance? An interventional ablation

Table I ablates the loss on the 7.66× GAN autoencoder (25k-step fine-tunes from a shared converged baseline). Each row differs from its comparator in exactly one factor: $\mathcal{L}^{\text{dec}}_{\text{mix}}$ and $\mathcal{L}^{\text{enc}}_{\text{mix}}$ against the baseline (the loss), $\mathcal{L}^{\text{dec}}_{\text{mix}}$+disc against $\mathcal{L}^{\text{dec}}_{\text{mix}}$ (the adversarial term), and frozen-encoder $\mathcal{L}^{\text{dec}}_{\text{mix}}$ against $\mathcal{L}^{\text{dec}}_{\text{mix}}$ (the encoder's freedom to adapt); the design is not a full factorial. Three observations drive the paper.

(1) Reconstruction is unaffected. SDR$_{\text{rec}}$ is flat across all configurations (11.2–11.4 dB), and so is FAD (0.044–0.046): the mixing loss buys equivariance for free, on both a sample-wise and a distributional measure of quality.

(2) The decode-mixing loss is the active ingredient; the discriminator on the mixed path is not. On the ground-truth metric, $\mathcal{L}^{\text{dec}}_{\text{mix}}$ alone improves SDR$^{\text{gt}}_{\text{lin}}$ from 8.0 to 10.2 dB and is the only run with MixRate < 1 (0.97). Adding an adversarial term to the mixed path ("+disc-on-mix") does *not* improve ground-truth equivariance (9.9 dB, MixRate 1.05), yet it dramatically inflates the decode-vs-decode SDR$_{\text{lin}}$ (12.1 → 16.0 dB). The discriminator pulls the two decode paths onto a common realistic manifold (our hypothesis), raising their mutual SI-SDR without moving either toward $\bar{x}$. This is direct evidence that SDR$_{\text{lin}}$ is gameable, and that given $\mathcal{L}^{\text{dec}}_{\text{mix}}$ no adversarial term is needed; whether the discriminator has an effect on its own, without $\mathcal{L}^{\text{dec}}_{\text{mix}}$, is not tested here.

Loss	SDR$_{\text{rec}}$	SDR$_{\text{lin}}$	SDR$^{\text{gt}}_{\text{lin}}$	ℓ_{lat}	MixRate	FAD
baseline (no mix)	+11.3	+12.0	+8.0	0.068	1.203	0.045
$\mathcal{L}_{\text{dec}}$	+11.4	+12.1	+10.2	0.052	0.972	0.044
$\mathcal{L}_{\text{dec}}$ + disc-on-mix	+11.4	+16.0	+9.9	0.050	1.052	0.044
$\mathcal{L}_{\text{enc}}$ (γ=5)	+11.3	+12.8	+8.4	0.044	1.175	0.044
$\mathcal{L}_{\text{enc}}$ (γ=10)	+11.3	+13.1	+8.5	0.039	1.165	0.044
$\mathcal{L}_{\text{enc}}$ (γ=20)	+11.3	+13.5	+8.7	0.034	1.153	0.045
$\mathcal{L}_{\text{dec}}$ frozen-enc	+11.2	+11.7	+9.5	0.072	1.056	0.046

TABLE I
MECHANISM ABLATION ON THE GAN AUTOENCODER (7.66×, FULL TEST SET, α=0.5). SDR$_{\text{lin}}$ IS DECODE-VS-DECODE; SDR$^{\text{gt}}_{\text{lin}}$ IS VS. GROUND TRUTH. SINGLE SEED.

(3) The effect is carried through the decoder. The encoder-only variant $\mathcal{L}^{\text{enc}}_{\text{mix}}$ drives ℓ_{lat} lowest but leaves MixRate high (1.15–1.18): it linearizes the encoder without making the decoder equivariant. Conversely, training only the decoder under $\mathcal{L}^{\text{dec}}_{\text{mix}}$ with the encoder frozen retains most of the gain (9.5 dB versus 10.2 joint and 8.0 baseline) while leaving encoder linearity unimproved (ℓ_{lat} 0.072). Joint training is best among the tested configurations.

B. Intervening on compression and architecture

Table II extends the recipe along two axes. **Compression:** trained *from scratch* at 15.3× (v3.1), the mixing model matches its no-mixing control (v3.0) on reconstruction (9.9 vs. 10.0 dB) while improving ground-truth equivariance by +1.7 dB (SDR$^{\text{gt}}_{\text{lin}}$ 6.8 → 8.6), MixRate (1.19 → 1.05), and encoder linearity (ℓ_{lat} 0.078 → 0.057). FAD, however, rises from 0.052 to 0.061 (Table II); whether that is perceptible is untested, so "no quality cost" is established at 7.66× (Table I) and only for SDR$_{\text{rec}}$ at 15.3×. Because v3.0 is a matched control (same architecture, compression, and schedule, mixing disabled), this isolates the loss's contribution and shows the effect is not an artifact of the looser 7.66× rate or of warm-starting. **Architecture:** we fine-tune the published Music2Latent [13] consistency autoencoder with the decode-mixing objective, the additivity supervision of [5] on its own architecture, using its native consistency loss on the mixed pair in place of the discriminator; this is the matched comparison in which only the decoder class differs from the GAN autoencoder. A continued-training control, the same fine-tune with the mixing terms disabled, moves equivariance by < 0.2 dB and *worsens* FAD, so the mixing model's gain is attributable to the loss, not to domain adaptation. Both decode-vs-decode and ground-truth metrics agree on the improvement, confirming it is not a measurement artifact. Absolute SDR$^{\text{gt}}_{\text{lin}}$ remains negative on Music2Latent because its consistency decoder is phase-incoherent with raw waveforms; the latent *geometry* improves (MixRate, ℓ_{lat}) but a consistency decoder does not convert this into clean waveform arithmetic as the GAN decoder does, so we report the GAN autoencoder as the primary result.

The gain is not domain-driven. Our test set spans three acoustically distinct domains: crowd-sourced indie production (FMA), solo piano (MAESTRO), and multi-track studio mixes

Model	$\mathrm{SDR}_{\mathrm{rec}}$	$\mathrm{SDR}_{\mathrm{lin}}$	$\mathrm{SDR}_{\mathrm{lin}}^{\mathrm{gt}}$	MixRate	FAD
GAN AE, 7.66×, no mix	+11.3	+12.0	+8.0	1.203	0.045
GAN AE, 7.66×, +mix	+11.4	+12.1	+10.2	0.972	0.044
GAN AE, 15.3×, no mix	+10.0	+10.6	+6.8	1.187	0.052
GAN AE, 15.3×, +mix	+9.9	+14.6	+8.6	1.046	0.061
M2L, no mix (baseline)	-3.3	+5.0	-9.6	1.101	0.127
M2L, fine-tune control	-3.1	+5.1	-9.5	1.125	–
M2L, +mix	-2.6	+6.9	-7.8	1.086	0.119

TABLE II
GENERALIZATION ACROSS COMPRESSION AND ARCHITECTURE (FULL TEST SET, α=0.5, EMA WEIGHTS FOR MUSIC2LATENT). THE M2L "FINE-TUNE CONTROL" IS THE CONTINUED-TRAINING BASELINE WITH MIXING DISABLED.

$\mathrm{SDR}_{\mathrm{lin}}^{\mathrm{gt}}$	FMA	MAESTRO	MUSDB	all
7.66×, no mix	+6.8	+13.8	+6.2	+8.0
7.66×, +mix	+8.9	+16.9	+8.2	+10.2
Δ	**+2.1**	**+3.1**	**+2.0**	**+2.3**
15.3×, no mix	+5.8	+12.0	+5.5	+6.8
15.3×, +mix	+7.2	+15.1	+6.7	+8.6
Δ	**+1.4**	**+3.2**	**+1.2**	**+1.7**

TABLE III
GROUND-TRUTH EQUIVARIANCE PER TEST DOMAIN (DB). EACH Δ IS AGAINST THE MATCHED NO-MIXING CONTROL AT THE SAME COMPRESSION RATE. THE GAIN IS PRESENT ON EVERY DOMAIN, NOT CARRIED BY ONE.

Model	drums	bass	vocals	other	all	gap↓
7.66× no mix	+4.2	+0.4	+5.2	+3.9	+3.4	5.2
$+f(0)$	+8.8	+6.8	+9.6	+8.0	+8.3	0.4
7.66× $+\mathcal{L}_{\mathrm{dec}}$	+5.7	+4.9	+7.2	+5.1	+5.7	3.0
$+f(0)$	+8.5	+6.9	+9.4	+8.3	+8.3	0.5
7.66× $+\mathcal{L}_{\mathrm{dec}}$+disc	+6.2	+5.4	+7.5	+5.0	+6.0	2.6
$+f(0)$	+8.5	+7.0	+9.3	+8.1	+8.2	0.4
15.3× no mix	+1.2	-1.6	+3.0	+2.1	+1.2	6.6
$+f(0)$	+7.3	+5.8	+8.8	+7.2	+7.3	0.5
15.3× +mix	+5.4	+4.7	+6.9	+3.9	+5.2	2.5
$+f(0)$	+7.3	+6.1	+8.5	+7.2	+7.3	0.4
M2L +mix	-9.9	-12.3	-9.9	-12.7	-11.2	8.4

TABLE IV
STEM REMOVAL VIA LATENT SUBTRACTION (SI-SDR, DB; MUSDB18 TEST). HIGHER IS BETTER; "GAP" IS THE LINEARITY TAX AGAINST EACH MODEL'S OWN ENCODE–DECODE CEILING (LOWER IS BETTER). "$+f(0)$" ROWS ADD THE ENCODED SILENCE TO THE SUBTRACTED LATENT (ORIGIN CORRECTION).

(MUSDB18). Table III breaks $\mathrm{SDR}_{\mathrm{lin}}^{\mathrm{gt}}$ out by domain against the matched no-mixing control at each compression rate. The improvement appears on all three at both rates ($+2.0$ to $+3.1$ dB at 7.66×; $+1.2$ to $+3.2$ dB at 15.3×), so it reflects a property of the learned latent space rather than an artifact of one recording style. The absolute ordering (MAESTRO easiest, MUSDB hardest) tracks reconstruction difficulty and is preserved by the mixing loss.

C. Downstream: latent-arithmetic stem removal

We remove a stem by latent subtraction, $\hat{x} = g(f(x_{\mathrm{mix}}) - f(x_{\mathrm{stem}}))$, and measure SI-SDR against the true residual on the MUSDB18 test set. This is an *oracle* setting: the removed stem is available and encoded, so it tests the representation's arithmetic, not blind separation; its use is latent-domain editing when stem codes exist, as in a pipeline that stores compressed stems or a latent generative model that composes sources without decoding each. Subtraction uses coefficients $(1, -1)$, which Eq. 1 does not cover (Section I), so we report raw subtraction and the origin-corrected form $g(f(x_{\mathrm{mix}}) - f(x_{\mathrm{stem}}) + f(0))$, exact for affine f, g. We also report the *linearity tax* gap $= \mathrm{SISDR}(g(f(x_{\mathrm{res}})), x_{\mathrm{res}}) - \mathrm{SISDR}(\hat{x}, x_{\mathrm{res}})$, the shortfall against the model's own encode–decode ceiling.

Raw subtraction favours the mixing loss; origin correction removes the difference. Raw, $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$ roughly doubles subtraction quality at 7.66× ($+3.4 \to +5.7$ dB) and the matched 15.3× pair goes from $+1.2$ to $+5.2$ dB. Adding $f(0)$, one encode of silence, lifts *every* model to within 0.5 dB of its ceiling (Table IV): $+2.0$ to $+6.0$ dB per model, on all 49 tracks. With the origin corrected the loss no longer helps subtraction: $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$ versus no mixing is -0.03 dB (95% track-cluster CI $[-0.15, +0.11]$) and the 15.3× pair $+0.01$ dB ($[-0.19, +0.21]$). The raw advantage was the no-mixing models' larger latent offset ($\|f(0)\|$ 6.6 versus 5.4 at 7.66×, 18.1 versus 14.8 at 15.3×), which coefficient-sum-zero arithmetic pays for. What the loss buys is convex equivariance (Tables I and III); subtraction needs the origin, a free post-hoc fix.

How much of this is measurement noise? Units from one recording are correlated, so the resampling unit is the track: a paired cluster bootstrap draws the 49 tracks with replacement (10^4 resamples). Raw contrasts sit far outside evaluation variance ($\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$: $+2.33$ dB $[+2.11, +2.54]$; 15.3×: $+4.02$ dB $[+3.71, +4.34]$; all 49 tracks), and the corrected contrasts are indistinguishable from zero, as above. Unit-level resampling gives intervals about three times narrower; we report track-level ones. They cover evaluation variance at the tested initialization, not variation across training runs.

D. Equivariance across mixing ratios

Sweeping α on a source-balanced subsample (Fig. 3), the mixing loss *flattens* the equivariance curve: the no-mixing baseline sags most at the hardest balanced mix (α=0.5) and recovers toward the extremes, while mixing models stay near-uniform across α (e.g. $\mathrm{SDR}_{\mathrm{lin}}^{\mathrm{gt}}$ swing of 1.2 dB for $\mathcal{L}_{\mathrm{mix}}^{\mathrm{dec}}$+disc vs. 3.2 dB for the baseline). Thus the property holds across mixing ratios, not only at the trained operating point.

We caution that the decode-vs-decode $\mathrm{SDR}_{\mathrm{lin}}$ is U-shaped in α for *all* models (near-identity mixes are trivially easy), which is a property of that metric, distinct from the phase-variance confound discussed in Section III-B.

V. CONCLUSION

A controlled study identifies an explicit decode-mixing loss as the training signal responsible for mixing-equivariance in a waveform autoencoder, obtained at no measured reconstruction cost. The loss itself, not an adversarial term on the mixed path, is the active ingredient for convex equivariance, and the effect transfers across compression rates. Raw stem

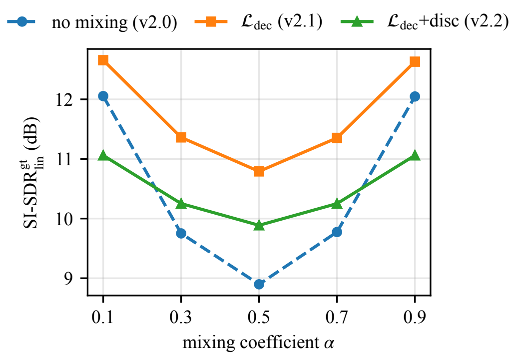

Fig. 3. Ground-truth equivariance vs. mixing coefficient α (source-balanced subsample). The no-mixing baseline sags at the hardest balanced mix (α=0.5); the decode-mixing loss flattens and lifts the curve across all α.

subtraction is governed by the latent offset, which one encode of silence removes for every model, with or without the loss. On a consistency-model autoencoder the loss improves latent geometry under a continued-training control without yielding usable waveform arithmetic, so practical claims are confined to the deterministic GAN decoder. **Limitations.** Every comparison is matched at a single initialization; this identifies the effect of each factor at that initialization but not its variation across training runs. Our confidence intervals cover evaluation variance (new tracks), so small differences (e.g. $\mathcal{L}^{\text{dec}}_{\text{mix}}$ vs. $\mathcal{L}^{\text{dec}}_{\text{mix}}$+disc, 0.31 dB) remain within plausible seed variance and we report them as "comparable." Experiments are mono; consistency-decoder latent arithmetic remains phase-limited; and we conduct no listening test. Stereo, multi-seed estimates, and latent-diffusion composition are future work.

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